Your LLM-judge RAG metrics are slower than the shortcuts that can replace them

The 8-hour test you didn't need to run — and the two metrics your judge can't actually judge.

If you build RAG systems, you've met the workflow: run RAGAS with an LLM-as-a-judge, wait hours, get a number, wonder if it means anything. I ran that workflow on a real agentic-RAG memory system — the session store of an LLM coding agent mirrored into a local vector database — and then asked a different question than "which metric is best":

> Which of these expensive judge metrics can a cheap, deterministic formula predict — and which ones is the judge itself getting wrong?

The results surprised me. Here's the short version.

What I did

- A local model (`gemma4:12b`) judged 100 real query/response pairs across the four classic RAGAS metrics. It took **8 hours and 26 minutes**.
- I then computed nine "shortcut" scores for the same samples — the kind of thing you can run in minutes on one GPU: sentence-level NLI for faithfulness, asymmetric QA embeddings for answer relevancy, graded NDCG with a cross-encoder reranker for context precision, and ROUGE-L + embedding cosine for answer correctness.
- I measured how well each shortcut predicts what the judge said (Spearman rank correlation over 99 joined samples).

What I found

1. AnswerCorrectness is replaceable — ρ = 0.89.

A blend of 75% ROUGE-L and 25% semantic cosine tracks the judge almost perfectly in rank. Same for ROUGE alone (0.87) and cosine alone (0.87). You can gate your CI on the cheap score and reserve the judge for occasional calibration runs.

2. AnswerRelevancy — strong with the right embedding (ρ = 0.71).

The same embedding family the retrieval index uses (`nomic-embed-text`) with a `search_query:/search_document:` prefix reproduces the judge's ranking strongly. A fine-tuned TAS-B encoder only got moderate agreement (0.58) and saturates numerically.

3. Faithfulness — the judge is lying (ρ = −0.18).

RAGAS gave a median score of **1.000** — everything is faithful! The NLI shortcut gave a median of **0.000** — nothing is entailed by the retrieved chunks! Two opposing verdicts, and the correlation is weakly *negative*. One of them is lenient, one strict, and neither can be trusted until you audit ~30 samples by hand.

4. ContextPrecision — no signal on this corpus (ρ ≈ 0).

Every variant was near-constant: bge-reranker NDCG ≈ 0.99. All retrieved chunks are relevant here, so the metric physically cannot rank anything. It's not that the metric is broken — it's that the corpus has no floor.

What this means for you

- **Run the shortcut-vs-judge check before you adopt any RAG metric on a new corpus.** It costs minutes and tells you which metrics are measuring something real.
- **Cheap beats slow for ranking.** Deterministic shortcuts don't drift, don't need a judge model loaded, and run in CI.
- **A zero correlation is a gift.** It tells you exactly which judge metric to distrust — and where to spend your audit budget.

The whole methodology, scripts, and aggregate tables are in the companion repository. No conversation content is shipped — only numbers, figures, and code.

Built entirely with local models: Chroma + nomic-embed-text + Ollama (gemma4:12b), PyTorch cross-encoders, and DeBERTa-v3 NLI.